

7-1 · Explore Integers and Their Opposites

Connect & apply · 8 tasks · Prepares for 6.NS.C.5

Use the explanation to discuss strategies. Accept mathematically equivalent answers.

1. FOUNDATION 1

On a number line marked in ones, what number is three steps right of 2?

5

Count 3, 4, 5 to the right.

Skill: Place whole numbers on a number line

2. FOUNDATION 2

Start at 2 and move backward 5 steps. Where do you land?

-3

Count 1, 0, -1, -2, -3.

Skill: Count forward and backward through zero

3. FOUNDATION 3

Write the integer for 7 degrees below zero.

-7

The negative sign indicates a value below zero.

Skill: Read a negative sign as “below zero” or “the opposite of”

4. FOUNDATION 4

Simplify $3x + 4x$.

$7x$

Add the coefficients of the like terms.

Skill: Simplify an algebraic expression carried in from Unit 6

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5. CHECK A

What is the opposite of -6 ? Where does -3 sit compared with 0?

Opposite of -6 is $+6$ (or 6). Position of -3 : sits 3 units to the left of 0 (or 3 units below 0).

Compare each part of your work with the worked solution.

6. CHECK B

What is the opposite of -9 ? Where does -5 sit compared with 0?

The opposite of -9 is 9. The number -5 sits 5 units to the left of 0.

Compare each part of your work with the worked solution.

7. CHECK C

What is the opposite of 12? Where does -1 sit compared with 0?

The opposite of 12 is -12 . The number -1 sits 1 unit to the left of 0.

Compare each part of your work with the worked solution.

8. TRANSFER

A diver is at -18 feet and a bird is at the opposite elevation. How far apart are they?

The bird is at $+18$ feet. The distance is $18 + 18 = 36$ feet.

Explain why your method works and include units when needed.

7-2 · Represent Rational Numbers and Their Opposites on the Number Line

Connect & apply · 8 tasks · Prepares for 6.NS.C.6.c

Use the explanation to discuss strategies. Accept mathematically equivalent answers.

1. FOUNDATION 1

Write the decimal halfway between 1 and 2.

1.5

Half of a unit is 0.5; add it to 1.

Skill: Place fractions and decimals between whole numbers

2. FOUNDATION 2

Between which two consecutive integers is -2.6?

-3 and -2

-2.6 lies to the right of -3 and to the left of -2.

Skill: Name the two consecutive integers a number falls between

3. FOUNDATION 3

Write $\frac{3}{5}$ as a decimal.

0.6

Convert $\frac{3}{5}$ to $\frac{6}{10}$, or divide 3 by 5.

Skill: Convert between a fraction and a decimal

4. FOUNDATION 4

An interval from 0 to 1 is split into 4 equal parts. What is the length of one part?

$\frac{1}{4}$

Divide the unit length into four equal shares.

Skill: Partition a number-line interval into equal parts

7-2 · Represent Rational Numbers and Their Opposites on the Number Line

Connect & apply · 8 tasks · Prepares for 6.NS.C.6.c

Use the explanation to discuss strategies. Accept mathematically equivalent answers.

5. CHECK A

Which two integers does 2.7 fall between? Where does $-1\frac{1}{2}$ go?

2.7 falls between integers 2 and 3. $-1\frac{1}{2} = -5.5$, which falls between -6 and -5 (exactly halfway between them).

Compare each part of your work with the worked solution.

6. CHECK B

Which two integers does 3.4 fall between? Where does $-7/2$ go?

3.4 falls between 3 and 4. $-7/2 = -3.5$, which sits halfway between -4 and -3 .

Compare each part of your work with the worked solution.

7. CHECK C

Which two integers does -1.2 fall between? Where does $9/4$ go?

-1.2 falls between -2 and -1 . $9/4 = 2.25$, which sits between 2 and 3, a quarter of the way up.

Compare each part of your work with the worked solution.

8. TRANSFER

Place $-2\frac{1}{4}$, -2.5 , and -2.75 in order from least to greatest.

-2.75 , -2.5 , $-2\frac{1}{4}$ (which is -2.25). Further left means less.

Explain why your method works and include units when needed.

7-3 · Understand Absolute Value of Rational Numbers

Connect & apply · 8 tasks · Prepares for 6.NS.C.7.c

Use the explanation to discuss strategies. Accept mathematically equivalent answers.

1. FOUNDATION 1

What is the distance from -4 to 0?

4

Count four unit intervals to zero.

Skill: Count the distance between two points on a number line

2. FOUNDATION 2

What is the opposite of -7?

7

Opposites are equally far from zero on opposite sides.

Skill: Name the opposite of a number

3. FOUNDATION 3

Evaluate the absolute value of -9.

9

Absolute value measures distance from zero, which is 9.

Skill: Report a distance as a positive number

4. FOUNDATION 4

A path goes 3 units to zero, then 5 more units. What total distance is traveled?

8

Add the two nonnegative distances: $3 + 5$.

Skill: Add two distances to find a total

7-3 · Understand Absolute Value of Rational Numbers

Connect & apply · 8 tasks · Prepares for 6.NS.C.7.c

Use the explanation to discuss strategies. Accept mathematically equivalent answers.

5. CHECK A

$|-9|$? How far is -4 from 0? How far from -4 to 3?

**$|-9| = 9$. Distance from -4 to 0 is 4 units.
Distance from -4 to 3: 4 units (to zero) + 3 units (to 3) = 7 units total.**

Compare each part of your work with the worked solution.

6. CHECK B

$|-12|$? How far is -7 from 0? How far from -7 to 2?

$|-12| = 12$. Distance from -7 to 0 is 7. From -7 to 2: $7 + 2 = 9$ units.

Compare each part of your work with the worked solution.

7. CHECK C

$|-3.5|$? How far is -6 from 0? How far from -2 to 8?

$|-3.5| = 3.5$. Distance from -6 to 0 is 6. From -2 to 8: $2 + 8 = 10$ units.

Compare each part of your work with the worked solution.

8. TRANSFER

Two numbers both have absolute value 6.5. What are they, and how far apart are they?

6.5 and -6.5 . They are 13 units apart.

Explain why your method works and include units when needed.

7-4 · Compare and Order Integers and Rational Numbers

Connect & apply · 8 tasks · Prepares for 6.NS.C.7.a

Use the explanation to discuss strategies. Accept mathematically equivalent answers.

1. FOUNDATION 1

Fill the blank with $<$ or $>$: -8 ___ -3 .

$<$

-8 is farther left on the number line, so it is less.

Skill: Use $<$ and $>$ correctly

2. FOUNDATION 2

Order -1 , 3 , -4 , 0 from least to greatest.

$-4, -1, 0, 3$

Read positions on the number line from left to right.

Skill: Place numbers on a number line

3. FOUNDATION 3

Which is greater: $\frac{3}{4}$ or 0.7 ?

$\frac{3}{4}$

Convert $\frac{3}{4}$ to 0.75 ; 0.75 is greater than 0.70 .

Skill: Compare decimals and fractions

4. FOUNDATION 4

Point A is at -5 and point B is at -2 . Which point is farther right?

B

-2 is to the right of -5 and is therefore greater.

Skill: Read left-is-less, right-is-greater on a number line

7-4 · Compare and Order Integers and Rational Numbers

Connect & apply · 8 tasks · Prepares for 6.NS.C.7.a

Use the explanation to discuss strategies. Accept mathematically equivalent answers.

5. CHECK A

-6 ____ -2 . Order these: -3 , 1.5 , $-1/2$, 0 .

$-6 < -2$ (since -6 is further to the left on the number line). Ordered least to greatest: -3 , $-1/2$, 0 , 1.5 .

Compare each part of your work with the worked solution.

6. CHECK B

-8 ____ -3 . Order these: -5 , 2.5 , $-1/2$, 0 .

$-8 < -3$. Order: -5 , $-1/2$, 0 , 2.5 .

Compare each part of your work with the worked solution.

7. CHECK C

-1 ____ -7 . Order these: -2 , 0.75 , $-3/4$, 3 .

$-1 > -7$. Order: -2 , $-3/4$, 0.75 , 3 .

Compare each part of your work with the worked solution.

8. TRANSFER

A temperature of -4°F rises 6 degrees. Is it now warmer or colder than -1°F ?

$-4 + 6 = 2^{\circ}\text{F}$, which is warmer than -1°F because 2 sits to the right of -1 .

Explain why your method works and include units when needed.

7-5 · Represent Rational Numbers on the Coordinate Plane

Connect & apply · 8 tasks · Prepares for 6.NS.C.6.b

Use the explanation to discuss strategies. Accept mathematically equivalent answers.

1. FOUNDATION 1

From the origin, move 2 right and 6 up. Give the ordered pair.

(2, 6)

The x-coordinate records horizontal movement; y records vertical movement.

Skill: Plot ordered pairs in Quadrant I

2. FOUNDATION 2

Which quadrant contains $(-3, 4)$?

Quadrant II

Negative x is left and positive y is up.

Skill: Read the sign of each coordinate to name a quadrant

3. FOUNDATION 3

Describe the moves from the origin to $(-4, -2)$.

4 left, 2 down

Move horizontally first; both signs are negative.

Skill: Move left or right first, then up or down from the origin

4. FOUNDATION 4

An axis counts by 5. What value is two tick intervals left of zero?

-10

Two intervals total 10 units; left of zero is negative.

Skill: Read a scaled axis before plotting

7-5 · Represent Rational Numbers on the Coordinate Plane

Connect & apply · 8 tasks · Prepares for 6.NS.C.6.b

Use the explanation to discuss strategies. Accept mathematically equivalent answers.

5. CHECK A

Which quadrant holds $(-4, 5)$? Plot $(3, -2)$.

$(-4, 5)$ has negative x and positive y , which is Quadrant II. Plot $(3, -2)$: Move 3 units right on x -axis, then 2 units down on y -axis (Quadrant IV).

Compare each part of your work with the worked solution.

6. CHECK B

Which quadrant holds $(-3, -6)$? Plot $(5, -1)$.

$(-3, -6)$ has both coordinates negative — Quadrant III. $(5, -1)$: 5 right, 1 down — Quadrant IV.

Compare each part of your work with the worked solution.

7. CHECK C

Which quadrant holds $(7, 2)$? Plot $(-2, 4)$.

$(7, 2)$ has both coordinates positive — Quadrant I. $(-2, 4)$: 2 left, 4 up — Quadrant II.

Compare each part of your work with the worked solution.

8. TRANSFER

Reflect $(-3, 4)$ across the y -axis, then across the x -axis. Where does it end up?

Across the y -axis: $(3, 4)$. Then across the x -axis: $(3, -4)$.

Explain why your method works and include units when needed.

7-6 · Determine Distance on the Coordinate Plane

Connect & apply · 8 tasks · Prepares for 6.NS.C.8

Use the explanation to discuss strategies. Accept mathematically equivalent answers.

1. FOUNDATION 1

Compute $6 - (-2)$.

8

The distance from -2 to 6 is $2 + 6 = 8$.

Skill: Subtract integers on a number line

2. FOUNDATION 2

Evaluate the absolute value of -12.

12

Distance from zero is nonnegative.

Skill: Use absolute value to report a distance

3. FOUNDATION 3

A plotted point is 3 units left and 5 units up from the origin. Give its coordinates.

(-3, 5)

Left gives negative x; up gives positive y.

Skill: Read the coordinates off a plotted point

4. FOUNDATION 4

Find the horizontal distance from (-4, 2) to (7, 2).

11

The segment crosses the y-axis: 4 units to zero plus 7 more units.

Skill: Add two distances when a segment crosses an axis

7-6 · Determine Distance on the Coordinate Plane

Connect & apply · 8 tasks · Prepares for 6.NS.C.8

Use the explanation to discuss strategies. Accept mathematically equivalent answers.

5. CHECK A

Distance from $(-5, 2)$ to $(3, 2)$? What is $|-5 - 3|$?

Distance from $(-5, 2)$ to $(3, 2)$: Segment is horizontal ($y=2$). Distance = $|-5 - 3| = |-8| = 8$ units (or 5 units to y -axis + 3 units to right = 8).

Compare each part of your work with the worked solution.

6. CHECK B

Distance from $(-6, 3)$ to $(2, 3)$? What is $|-6 - 2|$?

The segment is horizontal ($y = 3$). Distance = $|-6 - 2| = |-8| = 8$ units.

Compare each part of your work with the worked solution.

7. CHECK C

Distance from $(4, -5)$ to $(4, 7)$? What is $|-5 - 7|$?

The segment is vertical ($x = 4$). Distance = $|-5 - 7| = |-12| = 12$ units.

Compare each part of your work with the worked solution.

8. TRANSFER

Points $A(-3, 5)$ and $B(-3, -2)$. Find AB , then find a point C so that $BC = 4$ horizontally.

$AB = |5 - (-2)| = 7$ units. C could be $(1, -2)$ or $(-7, -2)$.

Explain why your method works and include units when needed.

7-7 · Represent Polygons on the Coordinate Plane

Connect & apply · 8 tasks · Prepares for 6.G.A.3

Use the explanation to discuss strategies. Accept mathematically equivalent answers.

1. FOUNDATION 1

Plot (0, 0), (4, 0), (4, 3), (0, 3) and connect in order. Name the shape.

Rectangle

Opposite sides are parallel and all four corners are right angles.

Skill: Plot several points and connect them in order

2. FOUNDATION 2

Find the vertical distance from (2, -3) to (2, 5).

8

The x-coordinates match; distance is $3 + 5 = 8$.

Skill: Find horizontal and vertical distances between vertices

3. FOUNDATION 3

Find the perimeter of a rectangle with side lengths 4 and 3.

14

Add all four sides: $4 + 3 + 4 + 3$.

Skill: Add side lengths to find perimeter

4. FOUNDATION 4

Find the area of a triangle with base 8 and height 5.

20

Half of the product 8×5 is 20.

Skill: Find the area of a rectangle or triangle

7-7 · Represent Polygons on the Coordinate Plane

Connect & apply · 8 tasks · Prepares for 6.G.A.3

Use the explanation to discuss strategies. Accept mathematically equivalent answers.

5. CHECK A

Plot (1,1), (5,1), (5,4), (1,4). What is the perimeter? The area?

Vertices form a rectangle: Horizontal length: $5 - 1 = 4$ units. Vertical height: $4 - 1 = 3$ units. Perimeter: $2(4 + 3) = 14$ units. Area: $4 \times 3 = 12$ square units.

Compare each part of your work with the worked solution.

6. CHECK B

Plot (2, 2), (8, 2), (8, 6), (2, 6). What is the perimeter? The area?

Length $8 - 2 = 6$, height $6 - 2 = 4$. Perimeter = $2(6 + 4) = 20$ units. Area = $6 \times 4 = 24$ square units.

Compare each part of your work with the worked solution.

7. CHECK C

Plot (−3, 1), (3, 1), (3, 5), (−3, 5). What is the perimeter? The area?

Length $3 - (-3) = 6$, height $5 - 1 = 4$. Perimeter = 20 units. Area = 24 square units.

Compare each part of your work with the worked solution.

8. TRANSFER

Three vertices of a rectangle are (1, 2), (7, 2), and (7, 9). Give the fourth vertex and the perimeter.

Fourth vertex (1, 9). Sides 6 and 7, so perimeter = $2(6 + 7) = 26$ units.

Explain why your method works and include units when needed.